

Q1. Genetic Algorithm

[Marks 6, 4+2, 15 min]

You are given the target string: "HELLO". You are given the initial population below: -

RGYXP,**HQLLA,****XELLO,****HZKLO**

A. Use a Genetic Algorithm to evolve the population of strings to match the target. **Describe** the steps of the GA process, including:

1. Initial population generation (given).
2. Fitness function (Evaluate as the number of characters matching in the correct position)
3. Selection (tournament selection select any two pairs and making sure good solutions go to next generation)
4. Crossover (Crossover point at 2)
5. Mutation (Fixed: The gene at index 3 in a selected chromosome is randomly replaced with any letter from a-z)
6. Termination condition

Note: Next population consists of 2 parents from the initial population and 2 offspring.

B. Show how the GA evolves toward the solution after the **first generations** and also **calculate the fitness value** for the evolved generation. Explain

Answer:

1. Initial Population Generation:

Example:

- RGYXP
- HQLLA
- XELLO
- HZKLO

2. Fitness Function:

Evaluate how close each individual string is to the target. Fitness can be defined as the number of characters matching in the correct position.

Example for target "HELLO":

- RGYXP → fitness = 0
- HQLLA → fitness = 2 (H and L)
- XELLO → fitness = 4 (E, L, L, O)
- HZKLO → fitness = 3 (H, L, O)

3. Selection:

Choose parents based on fitness (e.g., roulette wheel or tournament selection). Fitter individuals have a higher chance to be selected. Example: XELLO and HZKLO.

4. Crossover:

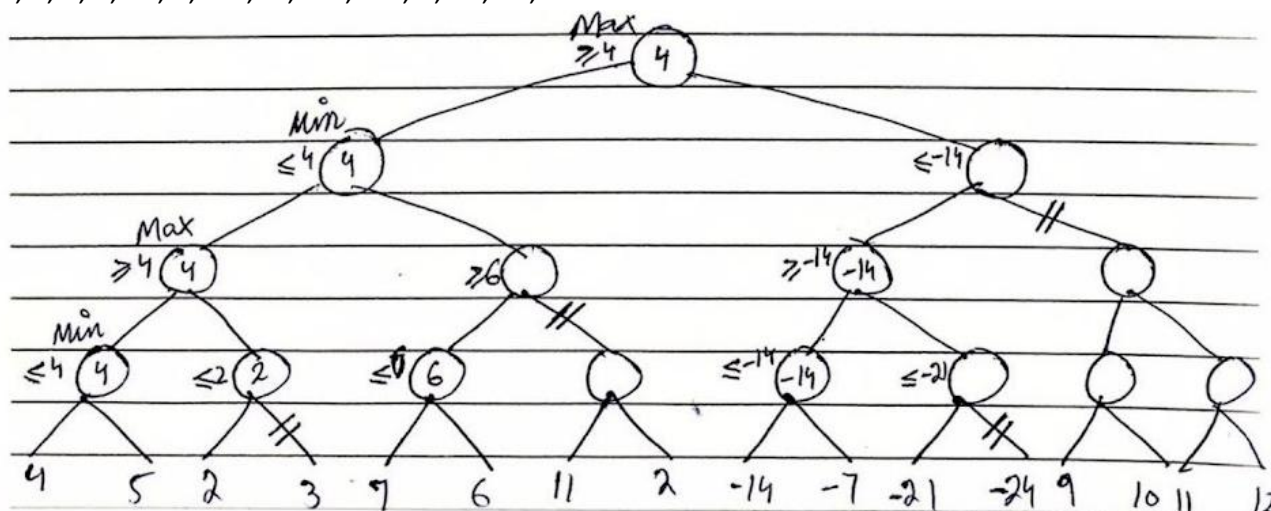
Combine parts of two parents to produce offspring.

Example crossover between XELLO and HZKLO:

- Crossover point at 2 → XE from parent 1 + KLO from parent 2 → XEKLO
- Or HZ from parent 2 + LLO from parent 1 → HZLLO

Randomly change a character with a low probability to introduce diversity.
Example mutation: XEKLO → HELLO (if X mutates to H, and K to L)

[Marks 4, 15 min]



[Marks 8, 4+4, 30 min]

2	$\beta < \alpha$ $\beta = \{a, b, c\}$ $\theta = \{a, b, c\}$	$\beta = \{a, b\}$ $\theta = \{a, b, c\}$	$\beta = \theta$ $\theta = \beta$ $\alpha > \beta$	New arc added in queue/Agenda	M1
3	$\beta = \theta$ $\beta = \{a, b\}$ $\theta = \{a, b, c\}$	No change	$\theta = \beta$ $\alpha > \beta$		M1 for updating Agenda
4	$\theta = \beta$ $\theta = \{a, b, c\}$ $\beta = \{a, b\}$	$\theta = \{a, b\}$ $\beta = \{a, b\}$	$\alpha > \beta$ $\beta = \theta$	New arc added in queue/Agenda	M1
5	$\alpha > \beta$ $\alpha = \{b, c\}$ $\beta = \{a, b\}$	No change	$\beta = \theta$		
6	$\beta = \theta$ $\beta = \{a, b\}$ $\theta = \{a, b\}$	No change	empty $\alpha = \{b, c\}$ $\beta = \{a, b\}$ $\theta = \{a, b\}$		A1 for correct Answer. Total = 5

B. You are scheduling 5 classes on Mondays, Wednesdays, and Fridays, with 3 professors where each professor can teach only one class at a time, and some classes have overlapping time slots.

Time	9:00 - 10:00	9:30 - 10:30	10:00 - 11:00	11:00 - 12:00	11:30 - 12:30
Classes	C1	C2	C3	C4	C5

Professor	Can teach	Other constraint
Ahmed (A)	C2, C4, & C5	
Babar (B)	C1, C2, & C4	
Chishti (C)	C1, C3, & C5	Refuses to teach 2 classes in a row

Draw the Constraint Graph for this problem.

Note: All classes are of exactly one hour duration.

Nodes C1, C2, C3, C4, C5

Edges C1-C2

time overlap C2-C3

C4-C5

M1

M1

M1

Chisti's constraint is not a time-Overlap constraint hence can not be represented as an "Edge", it is more like a sequential constraint.

Total = 3

Allow dotted line representation for Chisti's Constraint or just a suitable comment

If ignored/not shown -1

